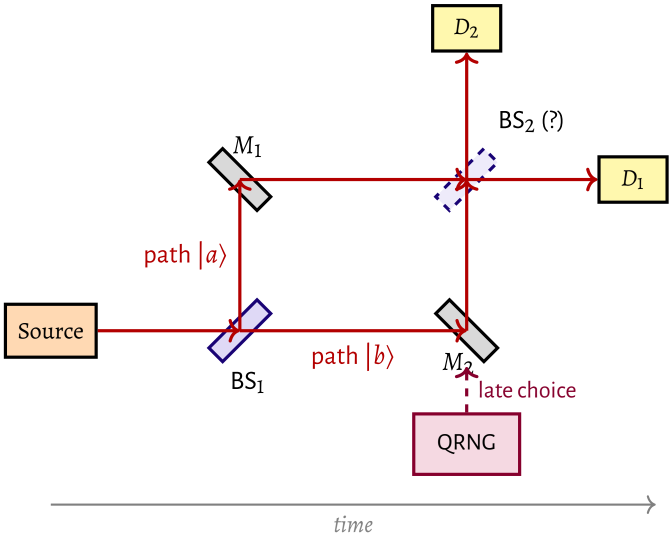

Figure 12 – **Schematic of the delayed-choice experiment (Mach–Zehnder version).** A photon emitted by the source strikes the beam splitter BS$_1$, creating a coherent superposition of the paths $|a\rangle$ (via mirror M_1) and $|b\rangle$ (via mirror M_2). **Crucial decision:** *after* the photon has already passed through BS$_1$ and is in the arms of the interferometer, a quantum random number generator (QRNG) decides whether the second splitter BS$_2$ will be present or absent. If BS$_2$ *is present*, the paths recombine and the photon exhibits *interference* (wave behavior) — detectors D_1 and D_2 register counts with probabilities depending on the relative phase between the arms. If BS$_2$ *is absent*, each detector is connected uniquely to one of the paths, revealing a *definite trajectory* (particle behavior). Wheeler's question was: how does the photon "know" which behavior to adopt if the decision was made after it had already entered the interferometer?